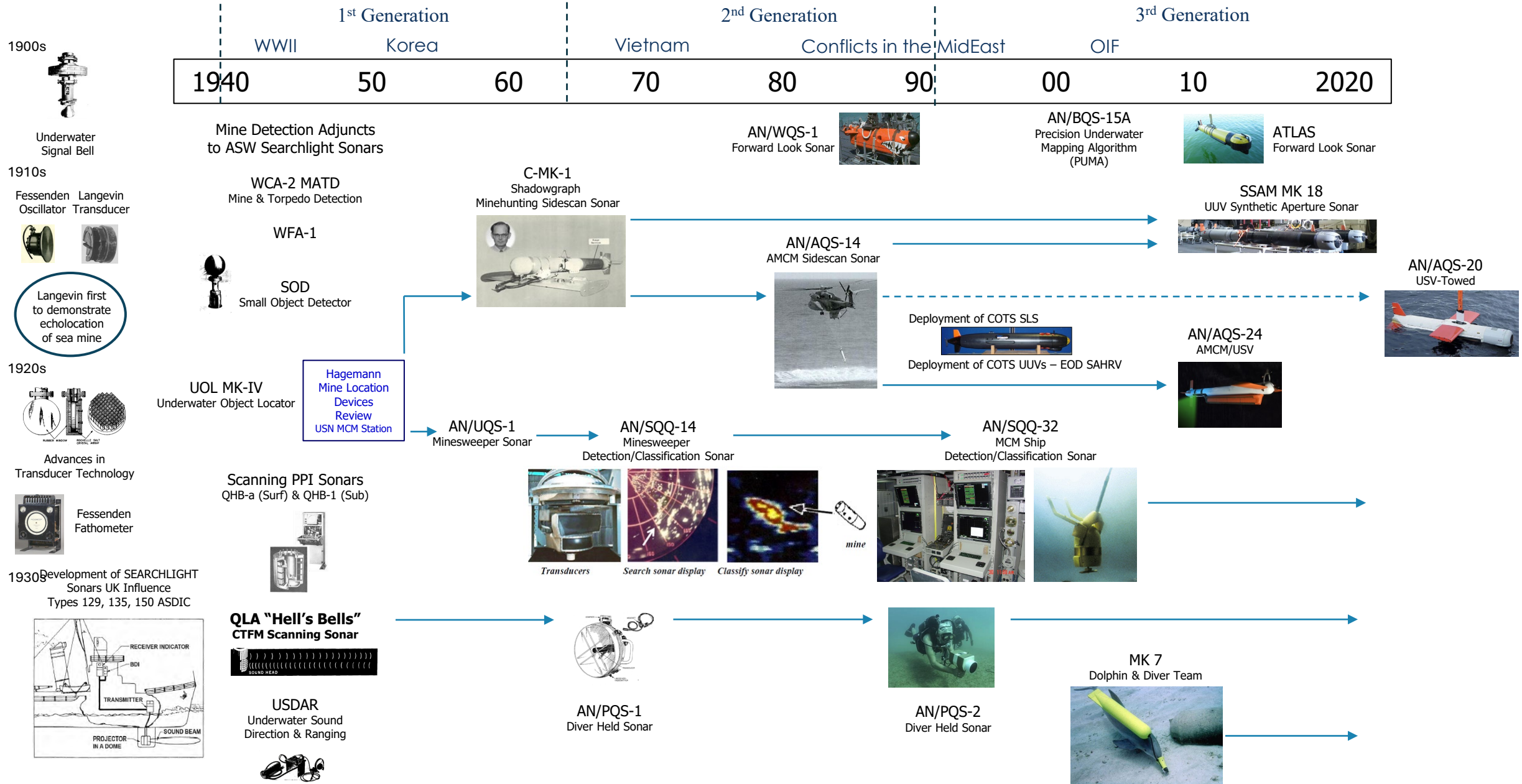


Evolution of Sonar and Sonar Signal Processing Fundamentals (Sample)

Dr. Spencer J. Chang
Dr. Daniel D. Sternlicht

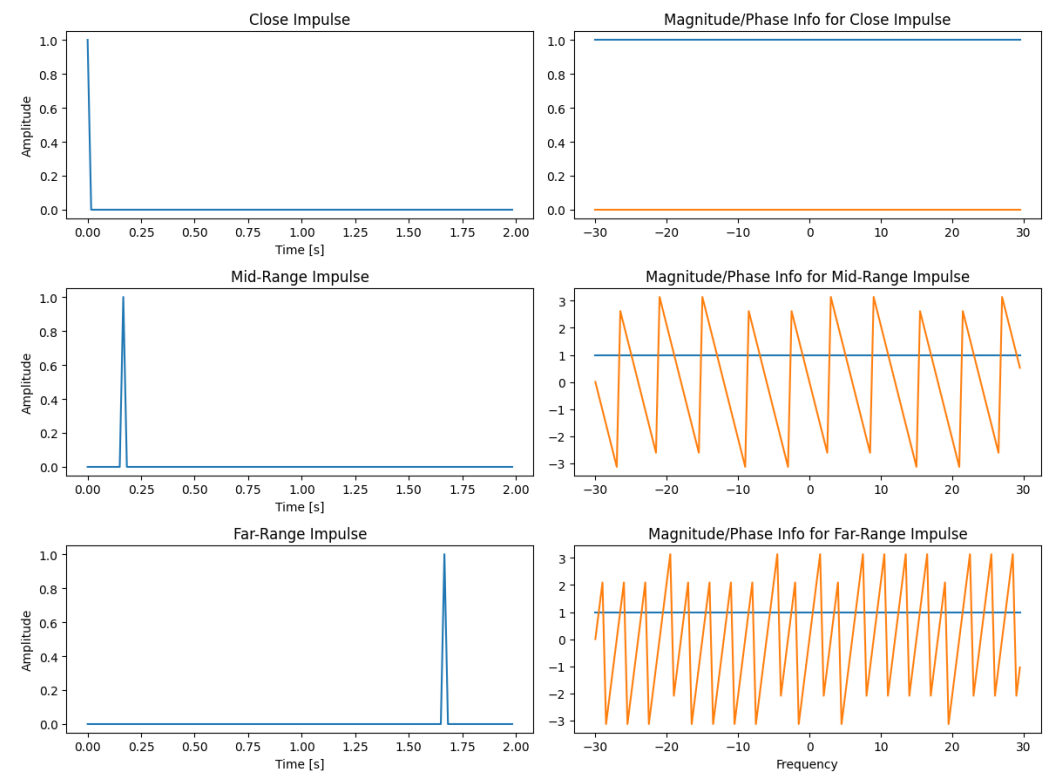
September 21, 2026

Evolution of Mine Detection Sonar (U.S.)

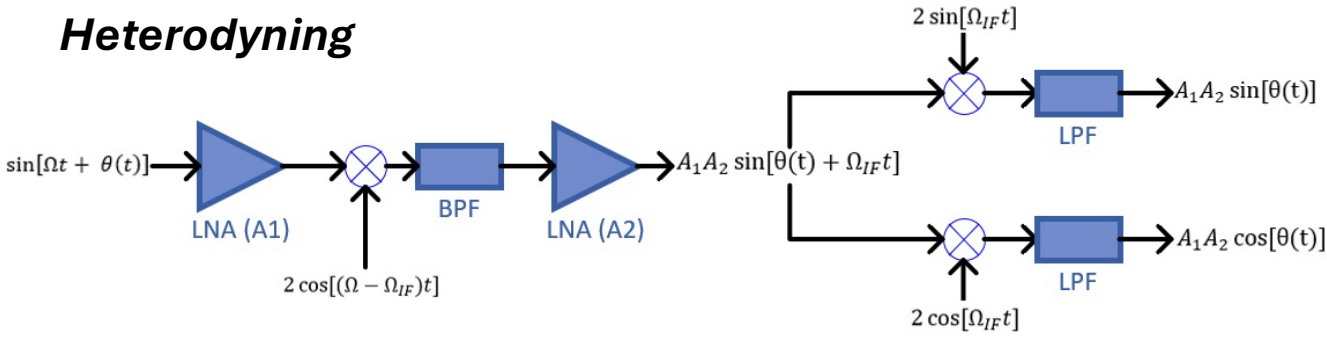


Signal Processing Fundamentals

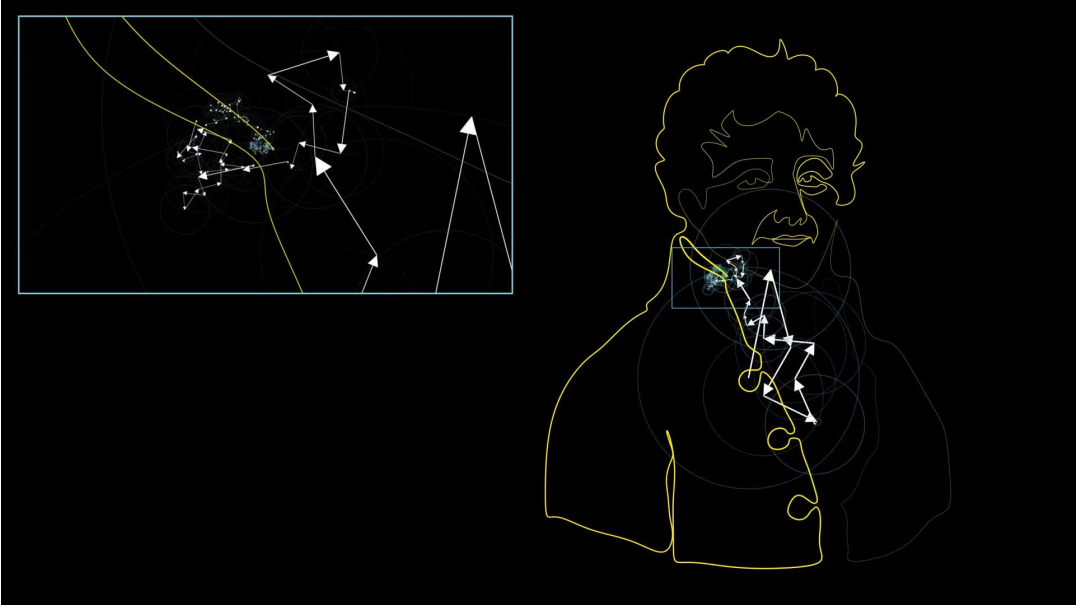
Time Delays and Phase Info.



Heterodyning

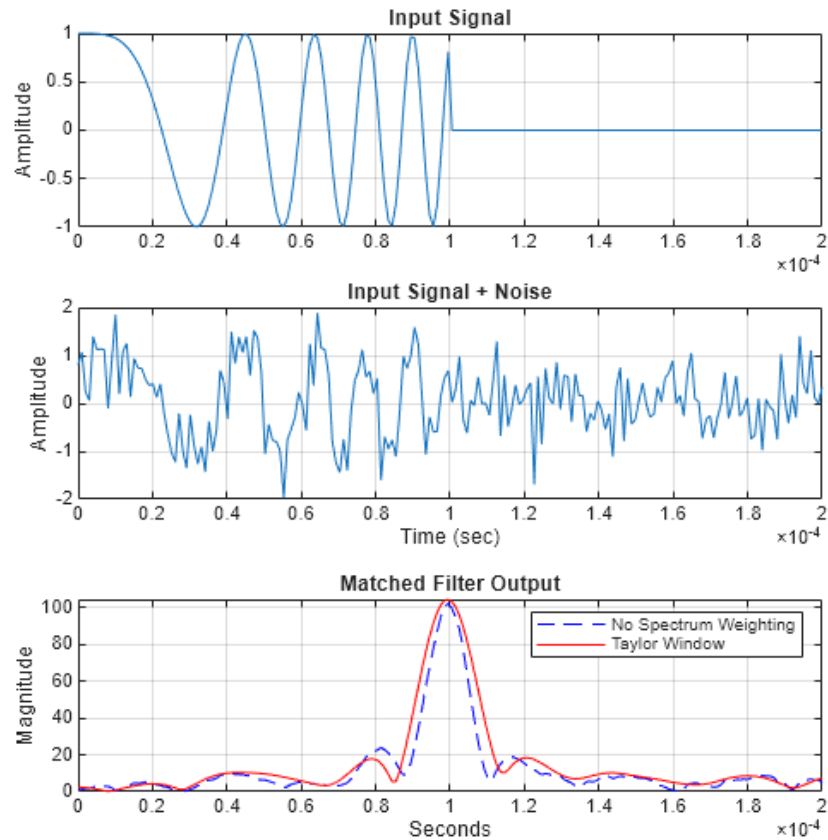


Fourier Analysis



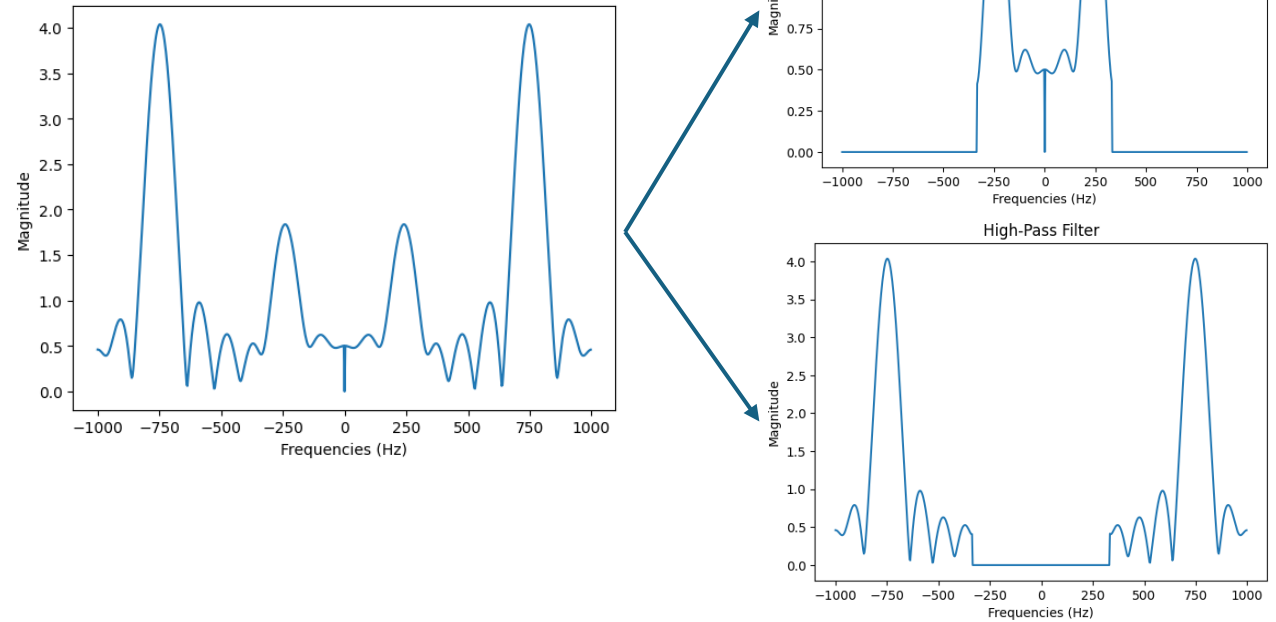
Filtering and Matched Filtering

Matched Filtering

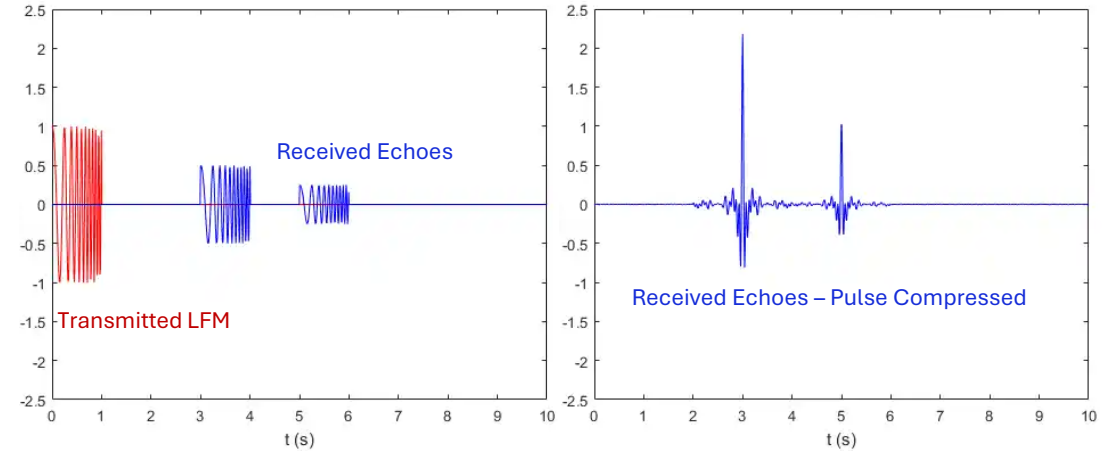


<https://www.mathworks.com/help/phased/ug/matched-filtering.html>

Basic Filtering



Pulse Compression



https://en.wikipedia.org/wiki/Pulse_compression

The Sonar Equation

Sonar equations are based on the “equality between the desired (i.e., the signal) and the undesired(i.e., the background) portions of the received signal...”

- RJ Urick, *Principles of Underwater Sound*

$$\textit{signal} = \textit{background}$$

Passive Sonar (only listen)

$$SL - TL = NL - DI$$

$$SNR_{\text{pass}} = SL - TL - (NL - DI)$$

SL – Source Level

TL – Transmission Loss

NL – Noise Level (ambient)

DI – Directivity Index

TS – Target Strength

RL – Reverberation Level

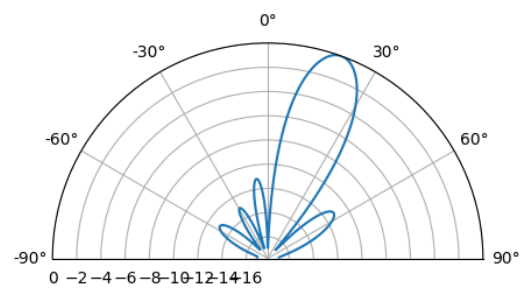
Active Sonar (project and listen)

$$SL - 2TL + TS = RL \oplus (NL - DI)$$

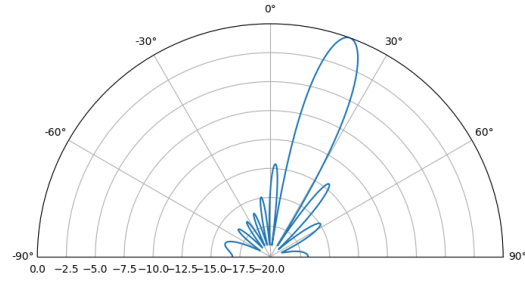
$$SNR_{\text{act}} = SL - 2TL + TS - RL \oplus (NL - DI)$$

DAS Beamforming: Linear Array

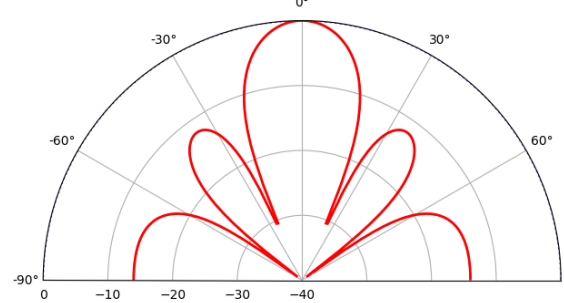
6 Receivers



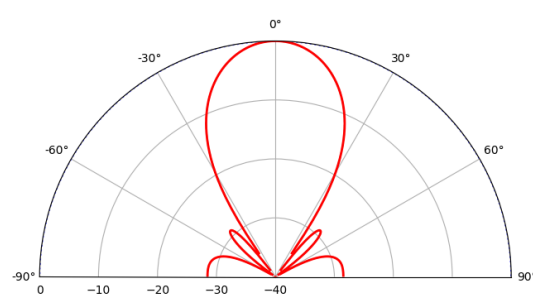
10 Receivers



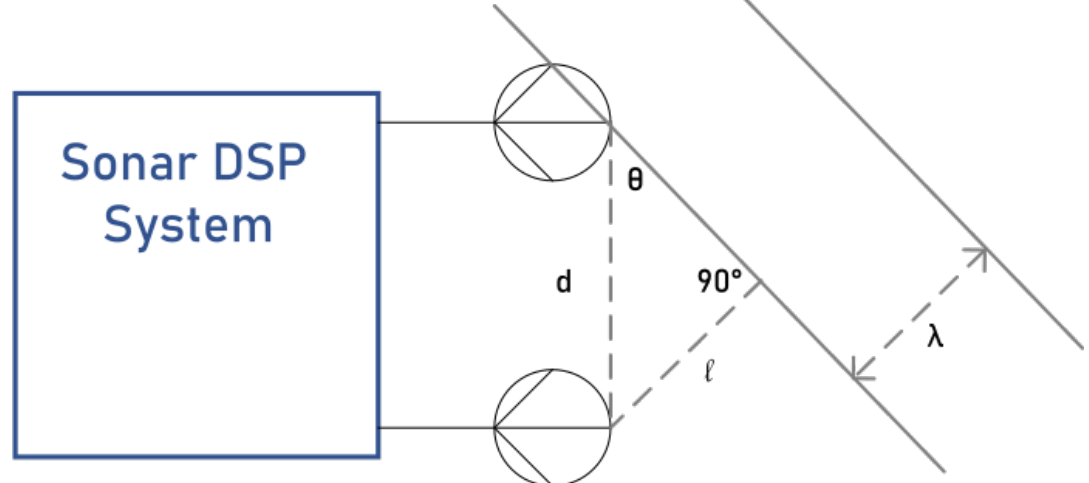
No Window



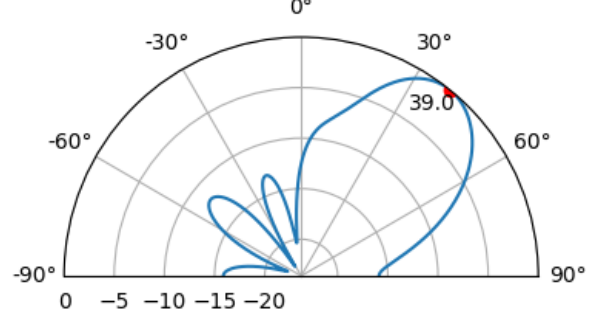
Taylor Window



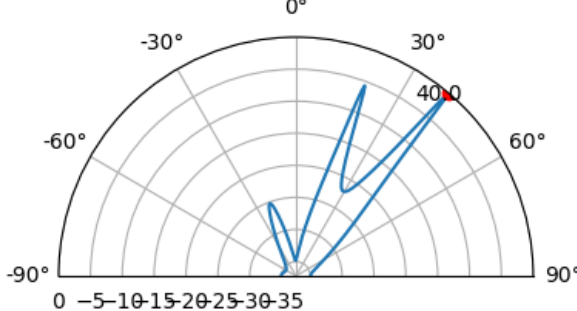
Multi-Element Receiver Array



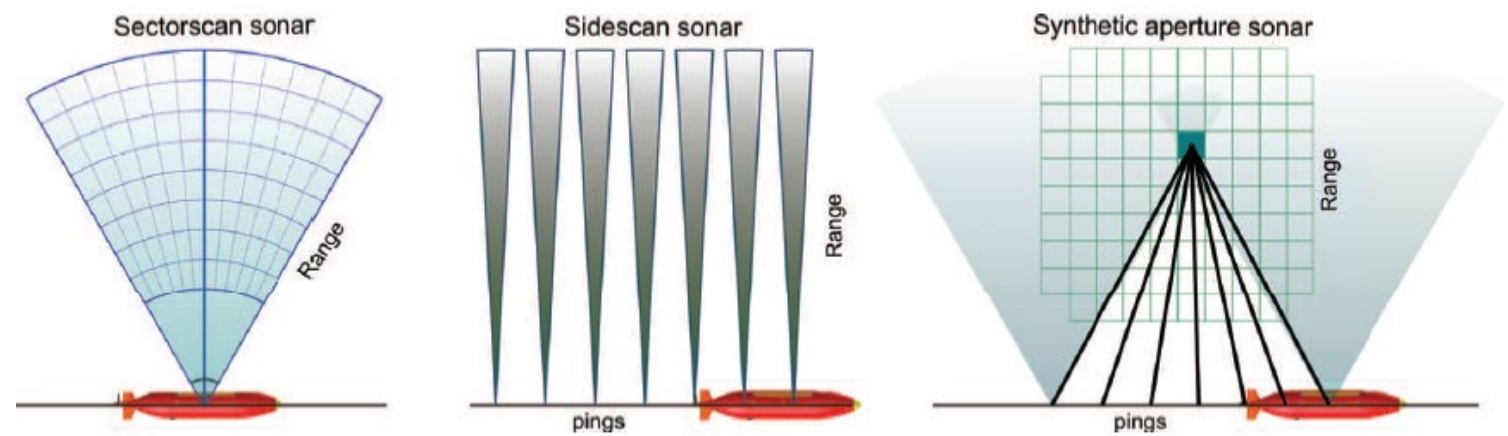
DAS Beamforming



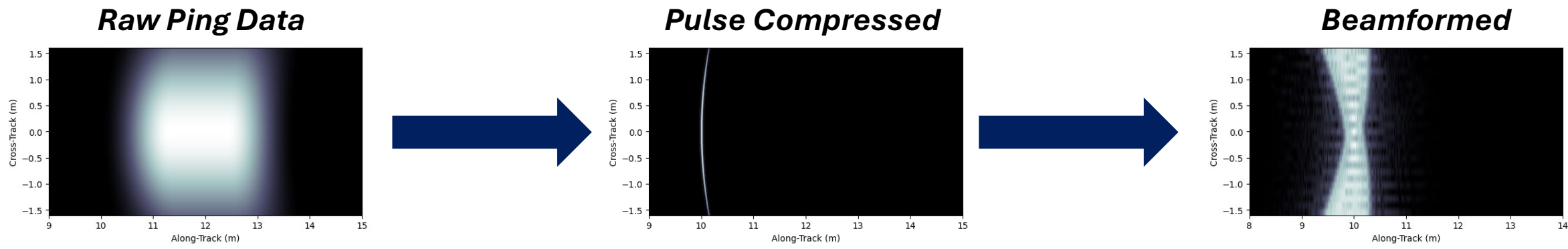
MVDR Beamforming



Sidescan Sonar and SAS Imagery



R. Edgar, 'Introduction to Synthetic Aperture Sonar', Sonar Systems. InTech, Sep. 12, 2011. doi: 10.5772/23122.



Data acquired from publicly available MASTODON tool: <https://github.com/Sonar-Sim/MASTODON>.